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ONCOLOGY

# Proton beam therapy for presumed and confirmed iris melanomas: a review of 36 cases

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**Abstract**
*Background* To report the clinical features and outcomes of iris melanomas treated by proton beam therapy.
*Materials and methods* A retrospective study was conducted at the Croix-Rousse University Hospital of Lyon, Department of Ophthalmology, in 36 patients treated by proton beam therapy for presumed ($n$=29) and confirmed ($n$=7) iris melanomas between July 1997 and October 2010. Ciliary body melanomas with iris involvement were excluded. The patients' mean age was 54.4 years (range, 22–82 years). The average tumor diameter was 3.8 mm (range, 2.5–8.0). The iridocorneal angle was invaded by the tumor in 47 % of cases ($n$=17), the ciliary body in 17 % of cases ($n$=6), and the sclera in 3 % ($n$=1). Raised intraocular pressure was present before treatment in 11.1 % of cases ($n$=4). Tumor biopsy was performed in 19 % of cases ($n$=7). Four patients had undergone an initial incomplete surgical excision of tumor before radiotherapy. Surgical preparation of the eye with tantalum ring positioning had been performed in all cases 3–4 weeks before irradiation. The prescribed dose was 60 Cobalt Gray Equivalent (CGE) of proton beam radiotherapy delivered in four fractions on four consecutive days.
*Results* The median follow-up was 50 months (mean 60.5, range 15–136). One patient (2.7 %) was lost to follow-up. None of the patients showed tumor progression, local recurrence, or metastasis. None of the patients required secondary enucleation. Cataract was developed in 62 % of patients, glaucoma in two cases (6 %) after irradiation, and hyphema with the aggravation of pre-existing glaucoma in one patient. No patients developed neovascular glaucoma.
*Conclusions* Proton beam therapy appears to be the treatment of choice for the conservative treatment of iris melanomas with excellent tumor control and an acceptable rate of complications. Longer follow-up studies on a larger series is necessary to consolidate these results.

**Keywords** Iris melanoma · Proton beam therapy · Irradiation · Tumor

## Introduction

Iris melanomas constitute 65–72 % of all of primary iris tumors but comprise only 4–10 % of all uveal melanomas [10, 13]. Identification of the clinical features suspicious of melanoma remains challenging. Several studies have reported that documented growth, basal diameter exceeding 3 mm, abnormal vasculature, pigment dispersion, satellite lesions, and tumor-related symptoms are highly suspicious of melanoma while ectropion uveae, anterior chamber angle involvement, slow growth, and even glaucoma do not reliably distinguish between naevi and melanomas [7, 22]. The various types of conservative treatment include surgical resection, irradiation, and a combination of these two approaches. Cases of radioactive plaque brachytherapy have been reported [21, 24]. Proton beam therapy is a modality of radiotherapy characterized by very precise dose delivery (Bragg peak) and a standard treatment for uveal melanomas. It was initially performed by B. Damato in Liverpool for the treatment of iris melanoma in 1994 [5, 12]. Since then, only a limited number

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of studies have reported the long-term outcomes of such an approach in the management of iris melanomas [5, 14, 18]. The aims of this study were to evaluate the results of proton beam radiotherapy of iris melanomas in terms of local tumor control, side effects, and ocular complications in a series of 36 consecutive patients seen over a 13-year period at the Croix Rousse University Hospital of Lyon, Department of Ophthalmology.

## Materials and methods

We extracted from our computerized diagnostic files all patients with clinically presumed and/or confirmed iris melanoma who were managed at the Croix-Rousse University Hospital of Lyon, Department of Ophthalmology and Biomedical Cyclotron of the Antoine Lacassagne Comprehensive Cancer Centre of Nice, between July, 1997 and October, 2010.

Informed consent for proton beam radiotherapy, with information on the risks and benefits of this treatment was obtained from the patients. The ethical review board of the University of Lyon approved the study.

Only patients treated for a primary iris melanoma were included. In agreement with other series, iris melanomas or lesions suspicious for iris melanoma were diagnosed clinically in the presence of an iris melanocytic lesion, and/or being >3 mm in diameter or 1 mm thick and/or being associated with one or more of the following features including prominent vascularity, ectropion uveae, secondary cataract, secondary glaucoma, and evidence of documented growth using serial slit-lamp photography and/or UBM (when available in our institution). Ciliary body melanomas with iris involvement were excluded. Patients with a follow-up of less than 12 months were excluded from the study.

*Patient characteristics* Thirty-six patients were treated, of which 24 were women and 12 were men. The mean age of patients was 54.4 years (range, 22–82 years). The tumor involved the right eye in 56 % of cases ($n$=20) and the left eye in 44 % ($n$=16) of cases. All lesions were unilateral. The iris color was blue in 56 % of patients ($n$=20), green in 17 % of patients ($n$=6), and brown in 28 % of patients ($n$=10).

Reasons for initial presentation are presented in Table 1. The mean interval between onset of the first symptom and the final diagnosis was 1.7 months (range, 0–30 months).

The mean visual acuity before the treatment was 20/25 (median, 20/20; range, 20/20–20/400). Two patients were pseudophakic before irradiation. Initial visual acuity worse than 20/50 was caused by cataract in four patients (11 %), corneal edema with hyphema in one patient (3 %), and age-related macular degeneration in one patient (3 %). The mean intraocular pressure was 15 mmHg (range, 10–40 mmHg) in the involved eye. Four patients (11 %) had an intraocular pressure greater than 21 mmHg, the highest being 40 mmHg, seen in a patient with hyphema and significant pigment seeding into the iridocorneal angle (Fig. 1). Other clinical features at initial assessment are listed in Table 2.

**Table 1** Reasons for initial presentation

| Reason for presentation | $n$ | % |
|---|---|---|
| Decreased vision | 4 | 11 |
| Allergic conjunctivitis | 1 | 3 |
| Acute anterior uveitis | 1 | 3 |
| Growth of a previously noted iris lesion | 14 | 39 |
| Photophobia | 1 | 3 |
| Ocular pain (related to raised IOP) | 1 | 3 |
| Spontaneous hyphema | 1 | 3 |
| Systematic follow-up of ocular melanocytosis | 1 | 3 |

In 26 (72 %) eyes, the tumor involved the inferior iris, with 12 (33 %) located in the infero-nasal quadrant, nine (25 %) located inferiorly, and five (14 %) inferotemporally. In nine (25 %) eyes, the tumor was located in the superior iris: five (14 %) superotemporally, two (6 %) superiorly, and two (6 %) superonasally. In one patient (3 %), there were five tumor nodules of melanomas on the same iris. Three patients (8 %) had the tapioca form of melanoma.

The mean clinical basal tumor diameter measured with the slit lamp was 3.8 mm (range, 2.5–8.0 mm). The tumor thickness measured by ultrasound biomicroscopy (UBM) in 14 patients (when available in our institution) ranged from 1 to 3 mm, with a mean of 2.5 mm. The mean number of clock hours of iris involvement was 2.6 (range, 1–8). Seventeen (47 %) tumors involved the iridocorneal angle, of which six (17 %) had clinical evidence of ciliary body involvement (Fig. 2). One tumor (3 %) had ciliary body and scleral involvement. None of the tumors showed extraocular extension.

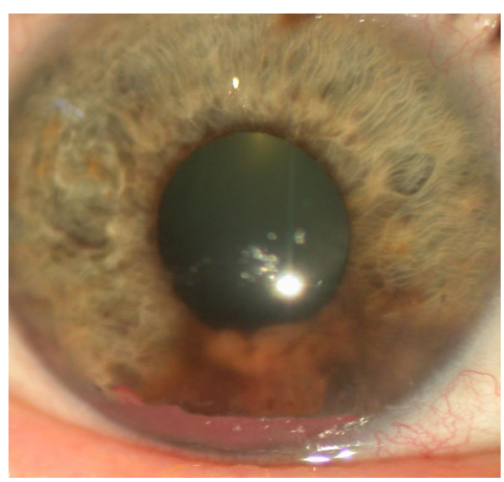


**Fig. 1** Growing tapioca iris melanoma in a 57-year-old woman associated with a hyphema and raised IOP (40 mmHg)

**Table 2** Clinical features at initial assessment

| Clinical features | *n* | % |
|---|---|---|
| Elevated IOP | 4 | 11 |
| Cataract | 4 | 11 |
| Hyphema | 1 | 3 |
| Ectropion uveae | 10 | 28 |
| Pupil distortion | 22 | 61 |
| Prominent intratumoral vascularization | 15 | 41 |
| Pigment dispersion | 18 | 50 |
| Iridocorneal angle involvement | 17 | 47 |
| Ciliary body involvement | 6 | 17 |
| Scleral extension | 1 | 3 |

*IOP* intraocular pressure, *n* number of patients affected, *%* percentage of patients

According to the 7th edition AJCC classification, there were 29 (81 %) tumors in T1, six (17 %) tumors in T2, and one (3 %) in the T3 size category. All patients were N0 M0 at the diagnosis.

A naevus was considered to precede the melanoma in 18 cases (50 %). Ocular melanosis was observed in one eye (3 %). In the remaining 17 cases, no pre-existing lesions were noted.

The melanoma was confirmed histologically in seven cases (19.4 %). In two patients, this was by fine-needle aspiration biopsy and in the rest by primary sectorial iridectomy. The resections were incomplete in all cases. One of the five cases had developed recurrent melanoma with five concurrent tumor nodules in the same iris, with ciliary and scleral involvement, 4 years after the primary sectorial iridectomy, which was performed at another ophthalmic centre. In this case, the primary reason for proton beam therapy was to salvage they eye. Histological examination showed the melanoma to be of spindle-cell type in six cases and of mixed spindle/epithelioid type in one case.

*Irradiation technique* The treatment procedure included the following: implantation under local anesthesia of 3–4 radio-opaque tantalum markers on the sclera to allow computerized, three-dimensional eye and tumor modeling and control of eye positioning during the treatment. A planning eye CT scan was used for eye, lens, and optic nerve delineation. The 3-D treatment-planning software "Eye Plan" dedicated to ocular tumors was used for eye modeling and planning of dose delivery in the tumor (Fig. 3).

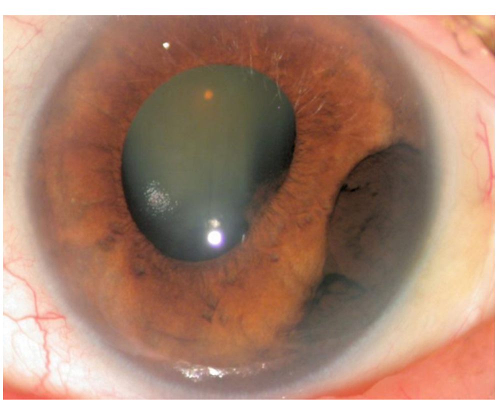


**Fig. 2** Documented enlarging iris melanoma involved the iridocorneal angle in a 77-year-old woman. Note the posterior synechia and cataract

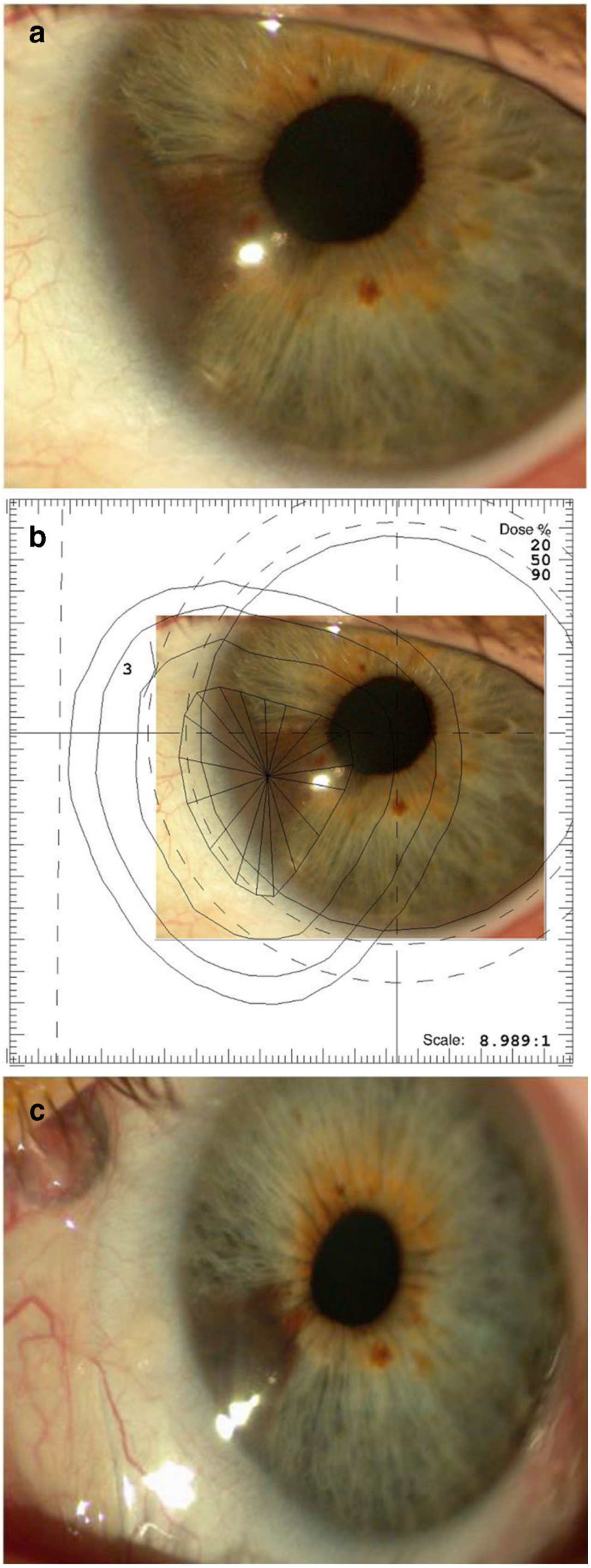
All patients were treated at Biomedical Cyclotron of the Antoine Lacassagne Comprehensive Cancer Centre (University Sophia Antipolis Nice) with a 65-MeV proton fixed horizontal beam line and the patient in a seated position. For head immobilization, a custom-made thermoplastic mask and dental impression (bite block) were used to ensure correct positioning of the eye. A fixation light was used to help the patient maintain the eye in the optimal gaze angle. Local anesthetic eye drops (Oxybuprocaine) were used at treatment when retracting the eyelids out of the radiation beam to avoid collateral damage. When total retraction of the eyelids was not possible (six patients or 16.6 %), the treatment was performed throughout the closed or half-closed eyelid.

Standard Polaroid orthogonal X-rays were replaced in 2008 by digital images for online check of the clip positions (i.e., eye position before each fraction). During the 10–20 s of beam-on treatment time, the eye position was monitored using an infrared camera (with images displayed on a monitor outside the treatment room) and the irradiation was paused if any eye movements occurred. This system allows control of eye positioning with an accuracy of 0.1 mm.

The beam range was usually selected to place the 90 % isodose of the distal fall 2.5 mm beyond the tumor. The 90 % isodose of the proximal spread-out-Bragg-peak (SOBP) was placed 2.5 mm in front of the tumor. The collimator border defined the location of the 50 % isodose. The dose delivered was 60 Cobalt Gray Equivalent (CGE) in four fractions of 15 Gy over 4 days [CGE is defined as the physical proton dose multiplied by the relative biological effectiveness (relative to Cobalt-60) of 1.1].

*Follow-up* Follow-up times were measured from the date of proton beam radiotherapy to the date of the event or the last known status.

All patients were reviewed at 1 month and 6 months after the proton beam radiotherapy, then 6 monthly for 5 years, and then annually. At each follow-up visit, examinations included the following: measurement of the best-corrected visual acuity (BCVA), slit-lamp examination, indirect ophthalmoscopy, and UBM when this became available in Lyon. In all patients, follow-up examinations at our center were alternated with assessments by the referring hospital. Screening for metastases consisted of 6-monthly hepatic ultrasonography. Outcome measures of these patients were reviewed retrospectively and the following outcomes were recorded: BCVA, presence of recurrent or persistent tumor growth, incidence and date of onset of




◀ **Fig. 3** Iris melanoma involving the peripheral iris and the irido-corneal angle. **a** Before treatment (October 1999). **b** Anterior proton field projection in an iris melanoma showing dose distribution around the tumor (20 %, 50 %, 90 % isodoses lines) using the Eye Plan software. **c** Regression of the tumor volume at 5 years after proton beam therapy (August 2004). Conjunctival hyperemia and sectorial iris atrophy developed as radiation side effects

ocular complications, such as cataract, hyphema, elevated IOP and its response to treatment, post-irradiation palpebral sequelae, dry eye syndrome, keratitis, posterior synechiae, uveitis and tantalum ring exteriorization. The tumor was classified as stable, partially regressed (i.e., reduction of tumor thickness), complete regressed (flat lesion), or showing progression (i.e., increased tumor thickness or diameter) (Fig. 3).

## Results

The median follow-up was 50 months (4.2 years) (range, 15–136). There were no immediate radiation-induced complications such as corneal abrasion or hyphema. Early radiation-related side effects occurring within the first month were all transient, and consisted of uveitis in one patient and corneal ulcer in two patients.

Two patients died of causes unrelated to the iris melanoma and one patient was lost to follow-up after 27 months. All other patients (33) were alive without metastasis at the close of the study. Local tumor control was achieved in all patients, with no cases of tumor progression or local recurrence. No patients died of metastasis. None of the patients required secondary enucleation for ocular complications. The mean of the latest BCVA was 20/32 (median, 20/25; range, 20/20–20/400). The causes of visual loss were: cataract with the patients waiting for surgery ($n$=5), glaucoma ($n$=1), band keratopathy ($n$=1), and pre-existing age-related macular degeneration ($n$=1).

The main post-irradiation complications are described in Table 3. Cataract was the first noted after the proton beam therapy in 25 patients. Overall, two patients were operated in the first year following proton therapy, no patients underwent

**Table 3** Ocular complications after proton beam therapy

| Complications | $n$ | % |
|---|---|---|
| Cataract post-irradiation | 25 | 62 |
| Glaucoma (new cases) | 2 | 6 |
| Hyphema | 1 | 3 |
| Recurrent corneal ulcer | 2 | 6 |
| Dystrophic corneal edema | 1 | 3 |
| Uveitis | 1 | 3 |
| Blepharitis | 1 | 3 |

Total patients: 36. Two were pseudo-phakic before irradiation cataract surgery in the second year, nine patients were operated on between the 3rd and 5th years after irradiation, and three patients were operated on between the 5th and 10th years after irradiation. In all cases, the phacoemulsification was used and there were no intraoperative incidents or post-operative complications. Glaucoma was present in four patients (11 %) before the treatment and developed in another two patients after the proton therapy. A variety of medications and surgical procedures, with varying degrees of success, were employed to manage the intraocular pressure. The glaucoma contributed to visual loss in one patient (3 %) after proton therapy.

Other complications and side effects after treatment included mild chronic blepharitis (one patient whose eyelids were included in the radiation field), hyphema (one patient), recurrent corneal ulcer (one patient, who had a pre-treatment band keratopathy), and peripheral localized corneal edema (1). Twenty-two patients (61 %) temporarily had a dry eye syndrome with conjunctival hyperemia and delayed conjunctival telangiectasia, which healed with topical lubricants. One patient developed sectorial iris atrophy. There were no patients developing scleral corneal necrosis, radiation retinopathy or papillopathy, definite rubeosis, or neovascular glaucoma.

## Discussion

The diagnosis of iris melanoma is highly likely if the tumor exceeds 1 mm in thickness, 3 mm in diameter, or if it shows diffuses spread, seeding, or growth. Ultrasound biomicroscopy (UBM) should allow better documentation and follow-up of iris lesions [1, 3, 13]. It is generally recommended that a melanocytic iris tumor can be initially observed every 3 or 6 months. If growth is documented or if tumor seeding or secondary glaucoma ensues, then treatment is suggested. In some doubtful cases, a fine-needle biopsy is indicated to confirm the diagnosis [2, 9, 11, 19, 27].

In this series, some patients showed growth of a tumor arising on a pre-existing lesion, while other patients presented with a tumor whose initial clinical characteristics were highly suggestive of the diagnosis of iris melanoma, in agreement with other series [5, 6, 16, 17, 26]. Biopsies confirming the malignant nature of the lesions were performed in 19 % of cases.

Several methods of conservative treatment of iris melanoma have been evaluated. Shields et al. administered plaque brachytherapy in a series of 14 patients with non-resectable iris melanoma. Complete regression of the tumor was achieved in 93 % of cases [24]. Proton beam therapy is an effective approach initiated by Damato in 1994 as an alternative to surgical excision [5], but this method remains limited only to certain centers. In our department, this therapeutic approach has been used increasingly since July 1997 as a first-line treatment instead of conventional sectorial iridectomy or irido-cyclectomy, even for small and circumscribed tumors of the iris. This is for the following reasons: (1) incomplete resection in some diffuse tumors (8 %), involvement of more than one quadrant of iris (6 %); (2) inability to resect the tumor in one case with multiple tumors, with ciliary body involvement and scleral extension (3 %); extension to the iris root (67 %) with iridocorneal angle involvement (47 %) and extensive ciliary body extension (17 %); (3) to minimize the side effects and post-surgical complications in the immediate and long term (i.e., photophobia, raised IOP, diplopia), which are more serious than those associated with proton therapy. Furthermore, there is a high incidence of recurrence after local excision.

The median follow-up in our series was 50 months. To the best of our knowledge, this is the longest follow-up of iris melanoma treated by proton beam therapy reported in the literature [5, 14, 18].

None of the patients of our series showed tumor progression or local recurrence. No patients developed iris neovascularization, despite irradiation of iris and ciliary body. We salvaged the eye in all cases and vision was preserved in most cases.

Local tumor recurrence is the main cause of secondary enucleation after proton therapy reported by some authors. In a series of 88 patients with iris melanoma treated by proton therapy reported by Damato et al., three eyes (3.4 %) were enucleated because of local tumor recurrence. The cumulative incidence of local tumor recurrence and enucleation at 5 years was 6 % (95 % CI, 0–14 %). In two patients, the recurrent tumor arose from untreated diffuse, circumferential spread around the anterior chamber angle. The third patient presented with an advanced tumor and received proton beam radiotherapy because she was reluctant to undergo primary enucleation. She developed recurrence 2 years after radiotherapy and died of metastasis 4 months later [5]. Rundle et al. reported one of 15 patients (6 %) with local recurrence of melanoma requiring enucleation after proton therapy, but no patients developed metastases over the period of follow-up [18].

The main complications were cataract (62 %, with a median follow-up of 50 months, excluding pre-treatment cataract). A third of cataracts (29 %) were operated on between the 3rd and 5th years after irradiation. Indeed, our data show good improvement in vision after cataract surgery in patients with no other causes of visual loss. Damato et al. [5] estimated a 63 % cumulative incidence of cataract at 4 years (95 % confidence interval [CI], 48–78 %). Using plaque radiotherapy for iris melanoma, Shields et al. reported a 70 % incidence of cataract at 5 years (6). The surgical management of radiation-induced cataracts was conventional and without major complications in all series reported in the literature [5, 14, 18].

Glaucoma was the second most frequent complication. Radiation-related glaucoma was noted in two patients (6 %). Proton therapy contributed to the worsening of a pre-treatment glaucoma in one patient, who developed visual loss. In all cases of our series, the glaucoma was of the open-angle type. There are no cases of iris rubeosis. Lumbroso-Le Rouic and Rundle had reported, respectively, rates of 15 % and 20 % of glaucoma after proton therapy of iris melanoma [14, 18]. The highest rate (33 % at 5 years) of post-irradiation glaucoma was reported by Shields in a series of 38 patients treated by $^{125}$I radioactive plaque radiotherapy. However, the tumors treated in this series were more extensive than the lesions of our series, with a median diameter of 9 mm (range, 4–13 mm) [21]. Except for the case of rubeosis iridis, the mechanism suggested was trabecular scarring [20, 25].

Hyphema associated with elevated IOP was noted in one case, and was easily treated with anti-glaucoma eye drops, without recurrence after 5 years of follow-up. Trichopoulos et al. have reported a case of recurrent hyphema after proton therapy for iris melanoma in which photodynamic therapy (PDT) was successfully used to treat this complication [28]. Corneal complications such as opacities or stromal dystrophies were rare and often peripheral, occurring in patients with a large tumor compressing the corneal endothelium. These complications seem to be more frequently reported in the series of iris melanomas treated by plaque radiotherapy [21, 24]. None of our patients required keratoplasty after proton beam therapy.

In four patients of our series, iris melanoma was initially managed by sectorial iridectomy and required proton therapy to achieve local control of residual tumor. The combination of these two therapeutic approaches, surgical resection followed by proton therapy, has rarely been described, but good results seem to be obtained [18].

The good survival in our patients is consistent with the published literature [4, 8, 15, 23]. Shields and associates reported a series of 169 patients with histologically proven iris melanoma managed by different methods. Metastases developed in 3 % of patients at 5 years, 5 % at 10 years, and 10 % at 20 years. The method of treatment (i.e., resection, radiotherapy, or enucleation) did not have an impact on metastasis. They conclude that the clinical factors at initial evaluation predictive of eventual metastasis from iris melanoma include: increasing age at diagnosis ($p$=0.03), elevated IOP ($p$=0.03), posterior tumor margin at angle or iris root (versus midzone) ($p$=0.02), extraocular extension ($p$=0.02), and prior surgical treatment of the tumor elsewhere before referral (versus observation) ($p$=0.006) [26].

In conclusion, our study provides further evidence that proton beam therapy is the treatment of choice for iris melanomas, with an excellent tumor control and acceptable side effects. Long-term follow-up and a larger series are necessary to consolidate these results.

**Disclosure information** The above authors confirm that this manuscript is not related to any proprietary or commercial interests. No sponsoring organizations have been involved, and no grants were received from any organization or institution.

**Ethics approval** This study was conducted with the approval of the Ethical Review Board of the University of Lyon.

**Meeting presentation** A limited account of this study was presented at the 36th Spring Meeting of the “Ophthalmic Oncology Group” (18th–20th March 2010, Liverpool, UK).

## References

1. Brown D, Boniuk M, Font RL (1990) Diffuse iris malignant melanoma of iris with metastases. Surv Ophthalmol 34:357–364
2. Char DH, Kemlitz AE, Miller T, Crawford JB (2006) Iris ring melanoma: fine needle biopsy. Br J Ophthalmol 90:420–422
3. Conway RM, Chew T, Golchet P, Desai K, Lin S, O’Brien J (2005) Ultrasound biomicroscopy: role in diagnosis and management in 130 consecutive patients evaluated for anterior segment tumors. Br J Ophthalmol 89:950–955
4. Conway RM, Chua WC-T, Qureshi C, Billson FA (2001) Primary iris melanoma: diagnostic features and outcome of conservative surgical treatment. Br J Ophthalmol 85:848–854
5. Damato B, Kacperek A, Chopra M, Sheen MA, Campbell IR, Errington RD (2005) Proton beam radiotherapy of iris melanoma. Int J Radiat Oncol Biol Phys 63:109–115
6. Demirci H, Shields CL, Shields JA, Eagle RC Jr, Honavar SG (2002) Diffuse iris melanoma: a report of 25 cases. Ophthalmology 109: 1553–1560
7. Geisse LH, Robertson DM (1985) Iris melanoma. Am J Ophthalmol 99:638–648
8. Harbour JW, Augsburger JJ, Eagle RC Jr (1995) Initial management and follow-up of melanocytic iris tumors. Ophthalmology 102:1987–1993
9. Hassenstein A, Bialasiewicz AA, Von Domarus D, Schäfer H, Richard G (1999) Tapioca melanomas of the iris: immunohistology and report on two cases. Graefe’s Arch Clin Exp Ophthalmol 237: 424–428
10. Henderson E, Margo CE (2008) Iris melanoma. Arch Pathol Lab Med 132:268–272
11. Inoue R, Saishin Y, Shima C, Yoshikawa H, Ohguro N, Tano Y (2009) A case of iris melanocytoma transformed to malignant melanoma. Jpn J Ophthalmol 53:271–273
12. Kacperek A (2009) Proton therapy of eye tumours in the UK: a review of treatment at Clatterbridge. Appl Radiat Isot 67:378–386
13. Kodjikian L (2003) Le mélanome de l’iris. In: Rigal D, Chiambaretta F, L’iris et sa pathologie. Rapport annuel. Bull Soc Opht France, Paris 95–115
14. Lumbroso-Le Rouic L, Delacroix S, Dendale R, Levy-Gabriel C, Feuvret L, Noel G, Plancher C, Nauraye C, Garcia P, Calugaru V, Asselain B, Desjardins L (2006) Proton beam therapy for iris melanomas. Eye (Lond) 20:1300–1305
15. Margo CE, Groden LR (1987) Iris melanoma with extensive corneal invasion and metastases. Am J Ophthalmol 104:543–544
16. Qian Y, Zakov ZN, Schoenfield L, Singh AD (2008) Iris melanoma arising in iris nevus in oculo (dermal) melanocytosis. Surv Ophthalmol 53:411–415
17. Reichert S, Berrod JP, Rozot P, Schmutz JL (1996) Melanoma developed on nevus of Ota without ocular melanosis. Apropos of an anatomo-clinical case. J Fr Ophtalmol 19:389–394
18. Rundle P, Singh AD, Rennie I (2007) Proton beam therapy for iris melanoma: a review of 15 cases. Eye (Lond) 21:79–82
19. Schalenbourg A, Uffer S, Zografos L (2008) Utility of a biopsy in suspicious pigmented iris tumors. Ophtalmic Res 40:267–272
20. Shields CL, Materin MA, Shields JA, Gershenbaum E, Singh AD, Smith A (2001) Factors associated with elevated intraocular pressure in eyes with iris melanoma. Br J Ophthalmol 85:666–669

21. Shields CL, Naseripour M, Shields JA, Freire J, Cater J (2003) Custom-designed plaque radiotherapy for non resectable iris melanoma in 38 patients: tumor control and ocular complications. Am J Ophthalmol 135:648–656
22. Shields JA, Sanborn GE, Augsburger JJ (1983) The differential diagnosis of malignant melanoma of the iris. A clinical study of 200 patients. Ophthalmology 90:716–720
23. Shields JA, Shields CL (1988) Hepatic metastases of diffuse iris melanoma 17 years after enucleation. Am J Ophthalmol 106:749–750
24. Shields CL, Shields JA, De Potter P, Singh AD, Hernandez C, Brady LW (1995) Treatment of non-resectable malignant iris tumours with custom designed plaque radiotherapy. Br J Ophthalmol 79:306–312
25. Shields JA, Shields CL, Kiratli H, de Potter P (1995) Metastatic tumors to the iris in 40 patients. Am J Ophthalmol 119:422–430
26. Shields CL, Shields JA, Materin M, Gershenbaum E, Singh AD, Smith A (2001) Iris melanoma: risk factors for metastasis in 169 consecutive patients. Ophthalmology 108:172–178
27. Sisley K, Brand C, Parsons MA, Maltby E, Rees RC, Rennie IG (1998) Cytogenetics of iris melanomas: disparity with other uveal tract melanomas. Cancer Genet Cytogenet 101:128–33
28. Trichopoulos N, Damato B (2007) Photodynamic therapy for recurrent hyphema after proton beam radiotherapy of iris melanoma. Graefe’s Arch Clin Exp Ophthalmol 245:1573–1575